

Computer Graphics Quick Reference Guide

Core Algorithms, Transformations, and Primitive Generations Explained

This reference document provides concise descriptions, mathematical frameworks, and analytical pros and cons for essential 2D/3D transformations, line, circle, and ellipse rasterization techniques, area-filling algorithms, and fundamental animation principles in Computer Graphics.

1. 2D Rotation

2D Rotation repositions an object along a circular path in a two-dimensional plane by a specified angular displacement θ around a designated pivot point. When rotating about the coordinate origin $(0,0)$ in a counter-clockwise direction, trigonometric functions map the source positions to new coordinates via matrix transformations.

$$\begin{aligned}x' &= x \cdot \cos(\theta) - y \cdot \sin(\theta) \\y' &= x \cdot \sin(\theta) + y \cdot \cos(\theta)\end{aligned}$$

ADVANTAGES (PROS)

- Enables precise, fluid angular orientation and orientation adjustments of 2D entities.
- Can be smoothly integrated into compound matrices via homogeneous coordinates alongside translation and scaling operations.

DISADVANTAGES (CONS)

- Requires significant computational overhead due to multiple floating-point evaluations of $\sin(\theta)$ and $\cos(\theta)$.
- Introduces precision aliasing and spatial rounding errors when fractional float coordinates drop down to discrete integer pixel grids.

2. 2D Translation

2D Translation alters the spatial location of an object by shifting it along a linear path from one position to another within a 2D plane. It uniformly applies rigid displacements t_x and t_y across all constituent coordinate points.

$$x' = x + t_x$$

$$y' = y + t_y$$

ADVANTAGES (PROS)

- Extremely lightweight performance overhead, relying entirely on basic integer/floating-point addition operations.
- Strictly preserves the foundational geometric shape, dimensions, and parallel orientations of structural bounds.

DISADVANTAGES (CONS)

- Incapable of modifying structural sizing, scale, or spatial orientation independently; must depend on complementary transforms for multifaceted operations.

3. 3D Scaling

3D Scaling updates the absolute physical size of a three-dimensional object by multiplying geometric coordinate matrices (x, y, z) by targeted scale multipliers (s_x, s_y, s_z) . Scaling is uniform when all parameters are identical, or non-uniform when values differ to distort dimensions along specific axes.

$$x' = x \cdot s_x$$

$$y' = y \cdot s_y$$

$$z' = z \cdot s_z$$

ADVANTAGES (PROS)

- Provides rapid sizing adaptations across three independent spatial dimensions using localized matrix modifications.
- Vital for enforcing perspective projection depth layers and specialized view adjustments.

DISADVANTAGES (CONS)

- Scaling relative to a default coordinate origin implicitly changes the object's global position, necessitating pre- and post-translation adjustments to perform local "in-place" scaling.

4. Boundary Filling

Boundary Fill is an area-coloring algorithm that fills an enclosed region bound by a uniform perimeter color. Initializing from an interior "seed" pixel position, it tests adjacent neighboring vectors (via 4-connected or 8-connected lookup orientations) and dynamically overlays pixels with a designated fill color until the perimeter boundary color threshold is struck.

ADVANTAGES (PROS)

- Highly intuitive conceptual layout and exceptionally straightforward execution pathways for complex, irregular geometric structures.

DISADVANTAGES (CONS)

- Extremely memory-intensive due to recursive stack dependencies, risking stack overflows across sprawling canvas frames.
- Vulnerable to color "leaks" if the border perimeter contains micro-fractures, incomplete vertex connections, or structural openings.

5. Bresenham's Line Generation Algorithm

Bresenham's Line Algorithm is a highly efficient rasterization framework that determines the precise pixel grid sequence required to approximate a continuous straight line between two endpoints. By checking step-by-step increments, it utilizes a localized incremental integer decision parameter to evaluate whether the true line path sits closer to the top or bottom pixel centers.

ADVANTAGES (PROS)

- Extremely fast operational performance; utilizes integer addition, subtraction, and bitwise shift math instead of complex floating-point computations or division.

DISADVANTAGES (CONS)

- Explicitly tailored to linear vectors; configuring the core framework to resolve non-linear curves, variable thickness, or sub-pixel anti-aliasing requires significant adaptations.

6. DDA (Digital Differential Analyzer) Line Drawing Algorithm

The DDA algorithm operates as a classic linear interpolation pipeline for rendering screen lines. By stepping across a primary driving axis by single-unit steps, it tracks corresponding fractional value changes along the secondary axis based on the explicit algebraic slope calculation $m = \Delta y / \Delta x$.

ADVANTAGES (PROS)

- Simpler to understand, implement, and mathematically trace compared to Bresenham's decision variable constraints.

DISADVANTAGES (CONS)

- Slower execution speeds stemming from frequent floating-point additions and explicit coordinate rounding calls (``round()``) at every iterative loop checkpoint.

7. Bresenham's (or Midpoint) Circle Drawing Algorithm

This method utilizes an optimization routine to render circular geometry on digital screens by taking advantage of 8-way circle symmetry. The underlying loop structure evaluates only a single 45-degree octant arc (0° to 45°) using an integer decision metric to toggle between horizontal or diagonal spatial shifts, then automatically projects matching points across the remaining 7 reflection octants.

ADVANTAGES (PROS)

- Incredibly fast execution profiles that eliminate floating-point arithmetic, geometric square roots, or trigonometric calls entirely.

DISADVANTAGES (CONS)

- Strictly bound to perfect geometric circles; cannot be adapted or morphed into asymmetric ellipse parameters without overhauling the inner decision variables.

8. Midpoint Ellipse Drawing Algorithm

Derived as an extension of the midpoint circle technique, this algorithm renders smooth, proportional ellipses by splitting structural processing across two distinct mathematical zones based on slope gradients (*slope* = -1 boundary). It leverages 4-way geometric reflection symmetry to project localized pixel arrays simultaneously across four quadrants.

ADVANTAGES (PROS)

- Maintains high rendering performance by avoiding costly square roots or float math during complex oval pixel mappings.

DISADVANTAGES (CONS)

- Highly complex implementation matrix due to the need to track, evaluate, and switch between two independent spatial step algorithms with separate decision formulas.

9. 2D Frame-by-Frame Animation of a Moving Car

This presentation architecture synthesizes horizontal movement by flashing separate pre-calculated graphical frames sequentially. At each timing loop checkpoint, the car body structure receives uniform translation updates while independent wheel primitives execute iterative angular rotations, combined with immediate clearing of the previous buffer frame data.

ADVANTAGES (PROS)

- Delivers absolute, fine-grained canvas state adjustments over asset dimensions, frame positioning, and interactive tracking configurations.

DISADVANTAGES (CONS)

- Prone to visible screen tearing, flickering, or motion stutter if structural rendering rates fall behind refresh rates, or if double-buffering is absent.

10. Scan-Line Polygon Fill Algorithm

This solid area filling algorithm processes polygons horizontally from top to bottom. It tracks intersection points between a virtual horizontal raster scan line and the structural lines of the polygon. By ordering intersections along an Edge Table (ET) and Active Edge Table (AET) from left to right, it shifts pixel color configurations to an "active" status for rows located between internal boundary pairs.

ADVANTAGES (PROS)

- Exceptional scaling parameters for processing complex, intersecting, or non-convex polygons cleanly with robust hardware-acceleration support.

DISADVANTAGES (CONS)

- Heavy memory architecture requirements and extensive algorithmic complexity due to continuous sorting, tracking, and edge management within active tables.

11. Smiley Face Generation

Smiley Face Generation serves as a practical application of combining multiple graphics primitives. It renders an outer head using the midpoint circle algorithm, maps two distinct smaller circle/ellipse variants for eye placement, and draws a curved structural arc equation or a clipped circle segment to render the smile boundary, relying on structural offsets to space elements relative to the head's center.

ADVANTAGES (PROS)

- Excellent composite foundational paradigm to demonstrate asset layering, local spatial coordinates, and multi-primitive group bindings.

DISADVANTAGES (CONS)

- Requires meticulous, hardcoded coordinate spatial tuning; scaling or translating the master shape requires updating the formulas for all inner components simultaneously.